Comparison of Modern Open-Source Multimodal LLMs (2025)

Scope

This document compares **modern, open-source, multilingual multimodal LLMs** that can process **text, images, and PDFs (via vision/OCR)**. The comparison focuses on **architecture, strengths, weaknesses, and best-fit use cases**.

Models Covered

1. **Llama-3.2-11B-Vision (Meta)**
2. **Llama-4-Scout-17B-16E (Meta)**
3. **Qwen2.5-VL-32B-Instruct (Alibaba)**
4. **GLM-4.6V (Zhipu / THUDM)**
5. **DeepSeek-VL2 (DeepSeek)**
6. **Pixtral-12B (Mistral AI)**
7. **Falcon-11B-VLM (TII)**
8. **Baichuan-Omni-1d5 (Baichuan)**
9. **Gemma-3-27B-IT (Google)**
10. **GPT-OSS-120B (OpenAI)****

High-Level Comparison

Model	Architecture	Multilingual	Vision Quality	PDF / Doc QA	Efficiency	License
Llama-3.2-11B-Vision	Adapter-based	⚠️Limited (Vision=EN)	Good	Good	🔲🔲🔲🔲	Custom Meta
Llama-4-Scout-17B-16E	MoE, Native	🔲🔲🔲🔲	Very Strong	Excellent	🔲🔲🔲🔲	Custom Meta
Qwen2.5-VL-32B	Dense	🔲🔲🔲🔲🔲	Industry-Leading	Excellent	🔲🔲	Apache 2.0
GLM-4.6V	MoE	🔲🔲🔲🔲	Very Strong	Excellent	🔲🔲🔲	Apache-style
DeepSeek-VL2	MoE	🔲🔲🔲🔲	Strong	Excellent	🔲🔲🔲🔲🔲	Apache 2.0
Pixtral-12B	Native multimodal	🔲🔲🔲🔲	Strong	Very Good	🔲🔲🔲🔲	Apache 2.0
Falcon-11B-VLM	CLIP-based	🔲🔲	Good	Fair	🔲🔲🔲	Falcon 2.0

Model	Architecture	Multilingual	Vision Quality	PDF / Doc QA	Efficiency	License
Baichuan-Omni-1d5	Omni-modal	⬜⬜⬜⬜	Good	Good	⬜⬜⬜⬜	Apache-style
Gemma-3-27B-IT	Native multimodal	⬜⬜⬜⬜⬜	Good	Very Good	⬜⬜⬜	Apache 2.0
GPT-OSS-120B	MoE (text-only)	⬜⬜⬜⬜	❌N/A	Text-only	⬜⬜⬜⬜	Apache 2.0

Model-by-Model Analysis

1. Llama-3.2-11B-Vision

Pros - Lightweight and efficient - Large 128k context - Excellent ecosystem support - Strong document VQA performance

Cons - Vision + text officially English-only - Vision is adapter-based (not fully native)

Best Use Cases - Edge / mobile deployment - Image captioning & DocVQA (English) - Lightweight multimodal assistants

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2. Llama-4-Scout-17B-16E

Pros - Native multimodal MoE architecture - Massive context (128k–1M+) - Strong reasoning across multiple images - Excellent multilingual performance

Cons - Newer ecosystem - More complex inference setup

Best Use Cases - Agentic systems - Long PDF + image reasoning - Enterprise knowledge assistants

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3. Qwen2.5-VL-32B-Instruct

Pros - Best-in-class vision & OCR - Excellent layout, chart, and table understanding - Strong multilingual support - Handles long videos & documents

Cons - Heavy (32B dense model) - Requires high VRAM

Best Use Cases - Enterprise document intelligence - Financial / legal PDF analysis - Visual agents & RPA

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4. GLM-4.6V

Pros - Native function calling - Excellent multimodal document understanding - Interleaved image + text generation - Strong Chinese & English support

Cons - Large full model (106B total) - Text-only reasoning slightly weaker than peers

Best Use Cases - Multimodal agents with tools - Screenshot-to-code / UI reasoning - Complex document workflows

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5. DeepSeek-VL2

Pros - Extremely efficient MoE design - Excellent OCR, tables, and charts - Strong visual grounding - Very cost-effective

Cons - Less conversational polish - Fewer community fine-tunes

Best Use Cases - Scalable document processing - Structured data extraction from PDFs - Cost-sensitive production systems

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6. Pixtral-12B (Mistral)

Pros - Natively multimodal (no adapters) - Apache 2.0 license - Strong text + vision balance - Variable image resolution support

Cons - Vision weaker than Qwen / GLM - No video or audio

Best Use Cases - General-purpose multimodal apps - Startups needing permissive licensing - RAG + PDF vision pipelines

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7. Falcon-11B-VLM

Pros - Good high-resolution perception - Permissive Falcon license - Lightweight deployment

Cons - Mostly English - Smaller context window - Weaker document reasoning

Best Use Cases - Research & experimentation - English-centric image QA - Academic environments

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8. Baichuan-Omni-1d5

Pros - True omni-modal (text, image, video, audio) - Real-time voice interaction - Strong medical image understanding

Cons - Smaller base model (7B) - Less suited for long-document reasoning

Best Use Cases - Voice assistants - Medical imaging - Real-time multimodal interfaces

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9. Gemma-3-27B-IT (Google)

Pros - Native multimodal (text + image) - Very strong multilingual coverage (140+ languages) - Long 128k context window - Apache 2.0 license

Cons - Vision weaker than Qwen / GLM on complex layouts - Smaller ecosystem than Llama / Qwen

Best Use Cases - Multilingual assistants - PDF understanding with OCR - Local / hybrid deployments

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10. GPT-OSS-120B (OpenAI)

Pros - Extremely strong reasoning & agent orchestration - Long-context (128k) text understanding - Native tool & function calling - Apache 2.0 license

Cons - Text-only (no vision or audio) - Requires OCR / vision model for PDFs

Best Use Cases - Agentic systems - RAG & workflow automation - Enterprise reasoning backends

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Practical Recommendations

Best Overall (Vision + PDFs)

- **Qwen2.5-VL-32B** (quality)
- **DeepSeek-VL2** (efficiency)

Best for Agents & Long Context

- **Llama-4-Scout-17B-16E**
- **GLM-4.6V**

Best Apache-Licensed Choice

- **Pixtral-12B**

- **DeepSeek-VL2**

Best Real-Time / Voice

- **Baichuan-Omni-1d5**

Best Multilingual Multimodal

- **Gemma-3-27B-IT**
- **Qwen2.5-VL-32B**

Best Text-Only Reasoning / Agents

- **GPT-OSS-120B**